



# Private Synthetic Locations

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**Series:** SYNTH | **Notebook:** 4 of 6 | **Created:** December 2025

## Monitoring Internal Applications from Your Infrastructure

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This notebook covers deploying and managing private synthetic locations (ActiveGates) for monitoring internal applications, APIs, and services not accessible from the public internet.

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## Prerequisites

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-  Access to a Dynatrace environment with Synthetic Monitoring
-  Completed SYNTH-01 through SYNTH-03
-  Infrastructure access to deploy ActiveGate (for setup)

# 1. Why Private Locations?

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## Public vs Private Locations

| Aspect      | Public Locations     | Private Locations   |
|-------------|----------------------|---------------------|
| Hosting     | Dynatrace cloud      | Your infrastructure |
| Access      | Public internet only | Internal networks   |
| Maintenance | Managed by Dynatrace | Managed by you      |
| Security    | External perspective | Behind firewall     |
| Latency     | Varies by region     | Local network       |

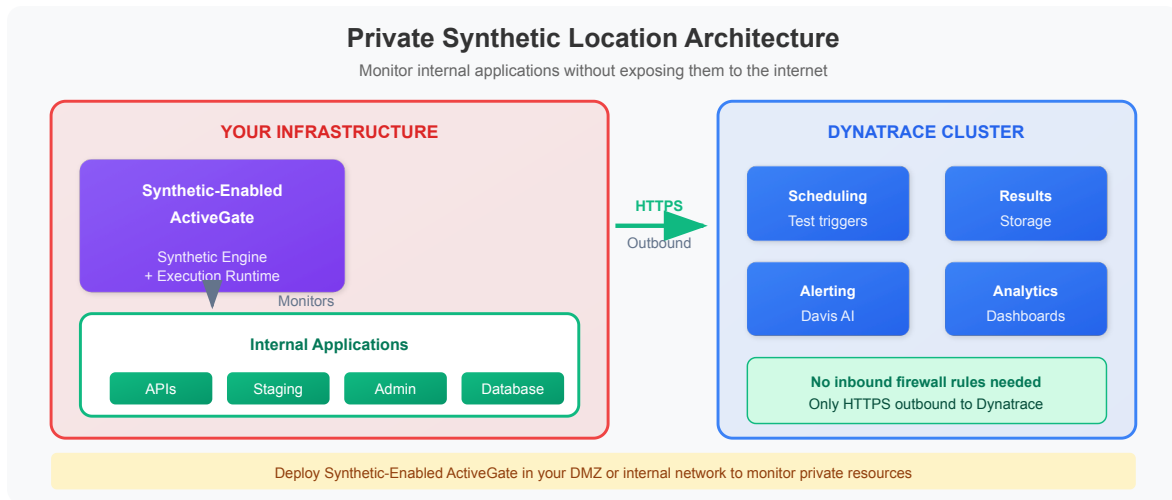
## Use Cases for Private Locations

| Scenario            | Description                       |
|---------------------|-----------------------------------|
| Internal APIs       | Services not exposed to internet  |
| VPN-only apps       | Applications requiring VPN access |
| Pre-production      | Staging/dev environments          |
| Security compliance | Data must not leave network       |
| Low latency testing | Local performance baselines       |
| Isolated networks   | Air-gapped or restricted networks |

# 2. Architecture

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Private synthetic locations use ActiveGates deployed within your infrastructure to execute monitors against internal applications:



## Key Points

- **Outbound only:** ActiveGate initiates all connections
- **No inbound ports:** No firewall changes for external access
- **Local execution:** Tests run inside your network
- **Results upload:** Only metrics/results sent to Dynatrace

## 3. Synthetic-Enabled ActiveGate

### ActiveGate Capabilities

ActiveGate can serve multiple purposes:

| Capability        | Description                |
|-------------------|----------------------------|
| <b>Synthetic</b>  | Private synthetic location |
| <b>Metrics</b>    | Metric ingestion endpoint  |
| <b>Logs</b>       | Log ingestion endpoint     |
| <b>API</b>        | Cluster API proxy          |
| <b>Extensions</b> | Extension execution        |

## Synthetic Engine Requirements

| Resource | Minimum        | Recommended            |
|----------|----------------|------------------------|
| CPU      | 2 cores        | 4+ cores               |
| RAM      | 4 GB           | 8+ GB                  |
| Disk     | 20 GB          | 50+ GB                 |
| Network  | HTTPS outbound | Low latency to targets |

## Browser Monitor Requirements

For browser monitors, additional requirements:

- Chrome/Chromium browser
- Display server (X11 or headless)
- Additional RAM for browser instances

## 4. Deployment Options

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### Option 1: Linux/Windows Installer

```
# Download from Dynatrace Hub
# Settings → Deployment status → ActiveGate

# Linux installation
sudo /bin/sh Dynatrace-ActiveGate-Linux-x86-*.sh \
  --enable-synthetic

# Verify synthetic capability
sudo systemctl status dynatracegateway
```

## Option 2: Container Deployment

```
# docker-compose.yml
version: '3'
services:
  activegate:
    image: dynatrace/activegate:latest
    environment:
      - DT_TENANT=abc12345
      - DT_CAPABILITIES=synthetic
      - DT_API_TOKEN=${DT_API_TOKEN}
    volumes:
      - activegate-data:/var/lib/dynatrace/gateway
    ports:
      - "9999:9999"
```

## Option 3: Kubernetes

```
# Use Dynatrace Operator with ActiveGate CRD
apiVersion: dynatrace.com/v1beta1
kind: DynaKube
metadata:
  name: dynakube
spec:
  activeGate:
    capabilities:
      - synthetic-monitoring
  resources:
    requests:
      cpu: "500m"
      memory: "512Mi"
```

```
// List all synthetic locations
// Note: 'type' (PRIVATE/PUBLIC), 'status', 'city', 'countryCode' fields are
// not available
// Private locations are typically identified by their naming convention
fetch dt.entity.synthetic_location
| fields id, entity.name
| sort entity.name asc
| limit 50
```

```
// Count synthetic locations
// Note: 'type' field not available on synthetic_location entity
// Use entity.name patterns to identify private vs public locations
fetch dt.entity.synthetic_location
| summarize {count = count()}
| fields count
```

## 5. Configuration

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### Creating a Private Location

1. **Deploy ActiveGate** with synthetic capability
2. **Navigate to:** Settings → Synthetic → Private synthetic locations
3. **Create location:** Name, description, geographic info
4. **Assign ActiveGates:** Select which ActiveGates serve this location

### Location Settings

| Setting                   | Description               |
|---------------------------|---------------------------|
| <b>Name</b>               | Descriptive location name |
| <b>Latitude/Longitude</b> | Geographic coordinates    |
| <b>City/Region</b>        | Location metadata         |
| <b>ActiveGate nodes</b>   | Assigned ActiveGates      |

### High Availability

For production workloads:

- Deploy 2+ ActiveGates per location
- Distribute across availability zones
- Load balancing automatic

```
// Monitor executions by location
// Note: Private locations typically have custom names (not city names like
"N. Virginia")
// Identify your private locations by their naming convention
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    executions = count(),
    availability_pct = round(countIf(synthetic.availability == true) * 100.0
/ count(), decimals: 2)
}, by: {dt.entity.synthetic_test, dt.entity.synthetic_location}
| sort executions desc
| limit 30
```

## 6. Monitoring Private Location Health

### ActiveGate Health Indicators

| Metric              | Description           | Alert Threshold |
|---------------------|-----------------------|-----------------|
| <b>Connectivity</b> | Connection to cluster | Any disconnect  |
| <b>CPU Usage</b>    | Processing load       | > 80% sustained |
| <b>Memory</b>       | RAM utilization       | > 80%           |
| <b>Disk</b>         | Storage usage         | > 80%           |
| <b>Queue Depth</b>  | Pending executions    | > 100           |

```
// Location availability over time
// Filter by your private location names using matchesPhrase if needed
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {
    success_count = countIf(synthetic.availability == true),
    total_count = count()
}, by: {dt.entity.synthetic_location, hour_bucket}
| fieldsAdd availability_pct = round((success_count * 100.0) / total_count,
decimals: 2)
| sort hour_bucket desc
```

```
// Execution count by location
// To filter for private locations, use entity.name patterns specific to your
naming convention
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total_executions = count(),
    failed_executions = countIf(synthetic.availability == false),
    unique_monitors = countDistinct(dt.entity.synthetic_test)
}, by: {dt.entity.synthetic_location}
| fieldsAdd failure_rate = round((failed_executions * 100.0) /
total_executions, decimals: 2)
| sort total_executions desc
```

## 7. Troubleshooting

### Common Issues

| Issue                      | Symptoms             | Resolution                     |
|----------------------------|----------------------|--------------------------------|
| <b>No connectivity</b>     | Location offline     | Check ActiveGate logs, network |
| <b>SSL errors</b>          | Certificate failures | Install CA certs on ActiveGate |
| <b>Timeouts</b>            | All tests fail       | Check network routes, DNS      |
| <b>Resource exhaustion</b> | Slow/failed tests    | Scale ActiveGate resources     |
| <b>Browser issues</b>      | Clickpath failures   | Check Chrome/display config    |

### ActiveGate Logs

```
# Linux log location
/var/log/dynatrace/gateway/

# Key log files
gateway.log      # Main gateway log
synthetic.log    # Synthetic execution log
connection.log   # Cluster connectivity
```



## Network Verification

```
# Test connectivity to target
curl -v https://internal-api.company.com/health

# DNS resolution
nslookup internal-api.company.com

# Test Dynatrace connectivity
curl -v https://&lt;tenant&gt;.live.dynatrace.com/api/v1/time
```

```
// Identify failing monitors by location
// To filter for specific private locations, add a filter on
dt.entity.synthetic_location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failure_count = count(),
    last_failure = max(timestamp)
}, by: {dt.entity.synthetic_test, dt.entity.synthetic_location}
| sort failure_count desc
| limit 20
```

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## Summary

In this notebook, you learned:

- ✅ **Why private locations** - Internal apps, security, compliance
  - ✅ **Architecture** - ActiveGate with synthetic engine
  - ✅ **Deployment options** - Installer, container, Kubernetes
  - ✅ **Configuration** - Creating and managing locations
  - ✅ **Health monitoring** - ActiveGate and execution metrics
  - ✅ **Troubleshooting** - Common issues and resolution
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## Next Steps

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Continue to **SYNTH-05: Network Monitoring** to learn about synthetic network availability monitoring.

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## References

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- [Private Synthetic Locations](#)
  - [ActiveGate Deployment](#)
  - [Synthetic-Enabled ActiveGate](#)
  - [ActiveGate in Kubernetes](#)
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